Across Layers: Auditing Symbolic and Acoustic Organization in a Pretrained Music Representation Model

Po-Chen Liu
Department of Computer Science
National Tsing Hua University
Hsinchu, Taiwan
noeltimliu@gamil.com

***Abstract*—Pretrained music representation models encode multiple aspects of musical signals, but a single notion of representation quality can obscure how different properties vary across model depth. This study audits MERT-v1-95M using two complementary views: controlled musical interventions and representational similarity analysis (RSA) on 30 deterministically selected MAESTRO v3 works. The pre-specified state-12 primary endpoint was positive (0.0799, 95% bootstrap CI $[0.0057, 0.1630]$), but the paired sign-flip test did not meet the conventional .05 threshold ($p = 0.0596$); the primary result is therefore non-confirming. Exploratory intervention profiles showed persistent sensitivity to the piano-to-harpsichord substitution and generally increasing pitch-class-versus-transposition separation toward later states, whereas symbolic RSA reached its descriptive maximum at state 8. Post-freeze robustness analyses further showed that threshold-based inference depends on the normalization/reference definition and finite sample composition. Together, the results show that *layer characterization is metric-dependent*: controlled intervention sensitivity and across-piece representational alignment capture different properties of the same representation hierarchy.**

***Index Terms*—music representation learning, MERT, representational similarity analysis, layer-wise analysis, music information retrieval**

I. Introduction

Large pretrained music-audio models such as MERT [1] have expanded the use of transferable representations in music information retrieval, while benchmarks such as MARBLE [2] emphasize that representation quality depends on the task used to evaluate it. A similar issue arises within a single pretrained model: describing one layer as "better" or "more musical" depends on what property is being measured.

A representation may respond strongly to a controlled change within a musical passage while showing a different pattern when its geometry is compared across different pieces. This study therefore separates two questions. First, how strongly do representations change under controlled musical interventions? Second, how closely does the geometry among original musical excerpts align with explicit symbolic and acoustic descriptor spaces?

The goal is not to identify a universally optimal layer. Instead, the study asks whether complementary audit measurements characterize model depth in the same way and how strongly such conclusions depend on the chosen measurement.

II. Methods

A. Sample

The formal sample consists of 30 unique works from MAESTRO v3, a paired audio–MIDI piano dataset introduced by Hawthorne *et al.* [3]. The works come from distinct composition families. One deterministic 20-second segment was selected per work using a quality-controlled procedure independent of embeddings and scientific outcomes. The sample is reproducible but is not probability-representative of the full corpus.

B. Controlled Interventions

Each intervention starts independently from the same original crop. Five conditions were retained:

1) Original;
2) Global transposition by $+2$ semitones;
3) Piano-to-harpsichord preset substitution;
4) Tempo increase from 120 to 150 BPM ($1.25\times$ faster);
5) Fixed octave-preserving pitch-class permutation.

The tempo condition retains its complete natural 16-second duration and therefore changes both tempo and input duration. The pitch-class intervention reassigns pitch classes while preserving timing, note duration, velocity, octave register, and other non-pitch symbolic attributes. It should therefore be interpreted as an operational probe of selected pitch relationships rather than a complete measure of symbolic musical structure.

The interventions were not calibrated to a common perceptual or acoustic magnitude. Accordingly, their absolute sensitivity values should not be interpreted as directly matched effect sizes.

C. Representation Extraction

MIDI was rendered using FluidSynth and the TimGM6mb SoundFont under a fixed synthetic rendering configuration. Audio was produced at 24 kHz and processed using the pinned MERT-v1-95M model [1] without training or fine-tuning.

All 13 returned states were retained: the input representation preceding the first transformer block and the outputs of the 12 transformer blocks. Whole-clip temporal mean pooling produced one 768-dimensional vector per state for each instance. The resulting formal dataset contains 150 instances: 30 works $\times$ 5 conditions.

D. Normalized Intervention Sensitivity

For work i, condition c, and state l, intervention distance is

$$d(i,c,l) = 1 - \cos\left(\mathbf{z}_{i,\mathrm{orig},l}, \mathbf{z}_{i,c,l}\right). \tag{1}$$

A same-state between-piece reference scale is defined as

$$b(i,l) = \mathrm{median}_{j \neq i}\left[1 - \cos\left(\mathbf{z}_{i,\mathrm{orig},l}, \mathbf{z}_{j,\mathrm{orig},l}\right)\right]. \tag{2}$$

Normalized sensitivity is then

$$s(i,c,l) = \frac{d(i,c,l)}{b(i,l)}. \tag{3}$$

This quantity is a reference-relative audit metric: it expresses the within-piece intervention distance relative to the typical same-state between-piece distance for that work. It is not an absolute perceptual or information scale.

The pre-specified primary endpoint at state 12 was

$$s_{\mathrm{PC}} - \frac{s_{\mathrm{transpose}} + s_{\mathrm{harpsichord}} + s_{\mathrm{tempo}}}{3}. \tag{4}$$

E. Representational Similarity Analysis

Representational similarity analysis (RSA) characterizes a representation by the pairwise dissimilarity structure induced over a common stimulus set [4], [5]. RSA used only the 30 original clips. At each state, a MERT cosine-distance representational dissimilarity matrix (RDM) was compared with two reference RDMs.

The symbolic descriptor contained normalized histogram blocks representing pitch class, onset position, note duration, and pitch height. The acoustic descriptor used log-mel temporal means and standard deviations. Spearman correlation between RDM upper triangles quantified alignment between MERT geometry and each descriptor geometry.

III. Results

A. Primary Endpoint

The state-12 primary estimate was 0.0799, with a percentile-bootstrap 95% confidence interval of $[0.0057, 0.1630]$. The two-sided paired sign-flip test produced $p = 0.0596$. The pre-specified test therefore did not meet the conventional .05 threshold, and the primary result is classified as non-confirming.

The positive percentile-bootstrap interval does not override this decision. The bootstrap interval and sign-flip test are distinct inferential procedures and are not inversion-equivalent; consequently, their threshold decisions need not coincide exactly in a finite sample.

TABLE I
Secondary contrasts at state 12.

Contrast	Mean	95% CI	Holm p
PC–Transpose	0.1394	[0.0895, 0.1914]	0.00029997
PC–Harpsichord	-0.4362	[-0.5622, -0.3030]	0.00029997
PC–Tempo	0.5365	[0.4542, 0.6276]	0.00029997

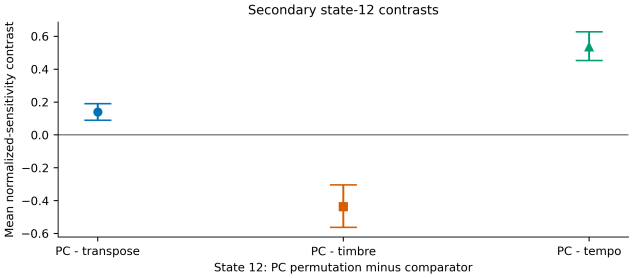


Fig. 1. State-12 secondary contrasts. Error bars show unadjusted percentile-bootstrap intervals. Multiplicity-controlled decisions use Holm-adjusted sign-flip tests.

B. Secondary State-12 Contrasts

Table I summarizes the three state-12 secondary contrasts, and Fig. 1 visualizes the paired differences. All three secondary null hypotheses were rejected after Holm adjustment. Pitch-class permutation produced greater normalized sensitivity than transposition and tempo, whereas the piano-to-harpsichord substitution produced greater sensitivity than pitch-class permutation. These contrasts do not replace the aggregate primary endpoint.

The exact secondary inferential procedure was fixed after descriptive directions had been inspected and before the secondary inferential computation.

C. Layer-Wise Intervention Profiles

The piano-to-harpsichord substitution produced the largest mean normalized sensitivity at every state. Its magnitude decreased substantially toward middle depth but remained prominent at later states. Pitch-class-minus-transposition separation generally increased with depth, although the trajectory contained local reversals.

The exploratory profiles in Fig. 2 therefore do not support a simple monotonic "deeper means more symbolic" interpretation. No layer-wise inferential procedure was used to select an optimal state.

D. Representational Similarity

Both symbolic and acoustic descriptor geometries showed positive alignment with MERT geometry across all states. Symbolic RSA reached its descriptive maximum at state 8 ($\rho = 0.6565$) and decreased to 0.6052 at state 12. The acoustic profile was high at early states, lower around middle depth, and recovered later; selected coefficients were 0.7172 at state

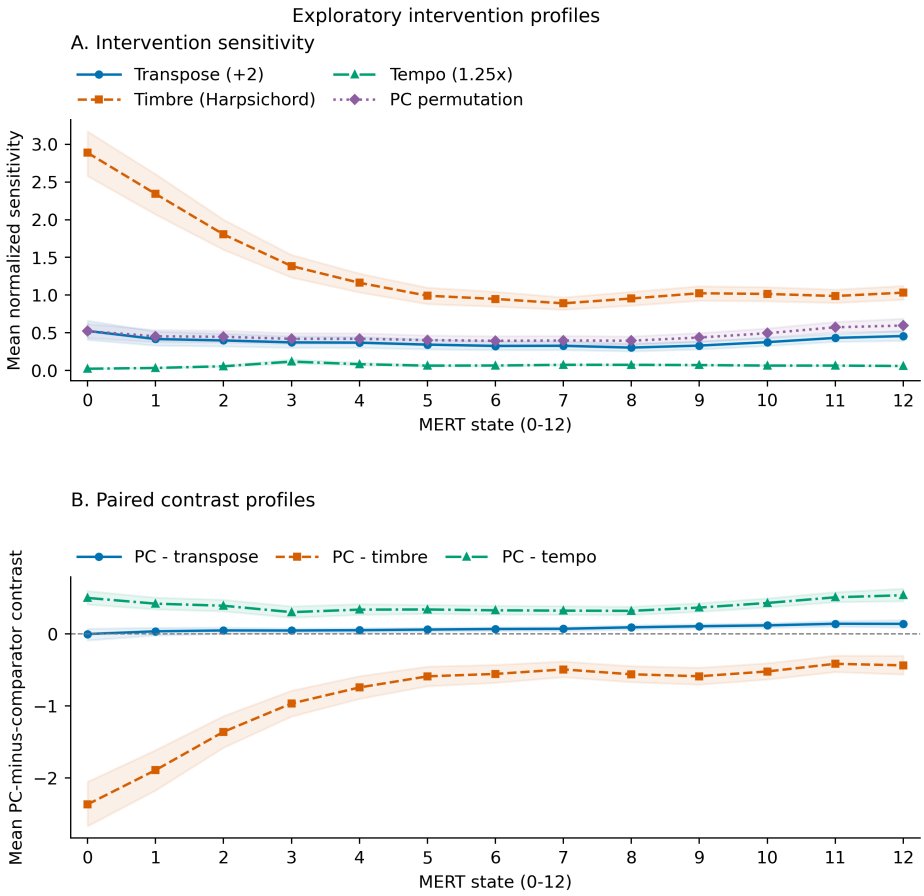


Fig. 2. Exploratory normalized-sensitivity profiles across all returned states. Intervals are descriptive and are not simultaneous layer-wise significance intervals.

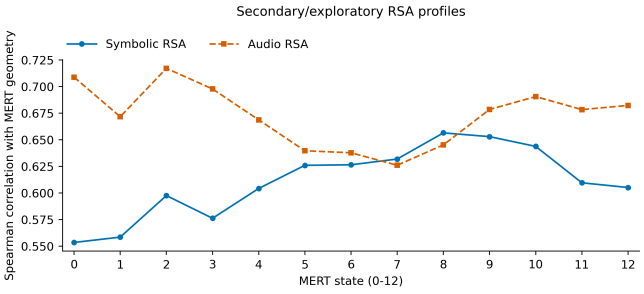


Fig. 3. Symbolic and acoustic RSA coefficients across model depth. The symbolic maximum at state 8 is descriptive rather than a statistically selected optimum.

2, 0.6261 at state 7, and 0.6824 at state 12. Figure 3 shows the two depth profiles.

These tests evaluate association between MERT geometry and each descriptor geometry. They do not test whether symbolic alignment is statistically stronger than acoustic alignment, nor whether state 8 is statistically superior to other states.

E. Post-Freeze Robustness Audit

After the scientific analysis had been frozen, additional robustness analyses were conducted and explicitly classified as post-freeze, post-hoc, and exploratory. The original normalized primary was first reproduced exactly before any alternative readout was examined.

Using raw cosine distances without the piece-specific normalization in Eq. (3), the state-12 contrast had mean 0.00747, 95% bootstrap CI $[0.00151, 0.01441]$, and a two-sided sign-flip $p = 0.0254$. This alternative result does not retroactively confirm or replace the pre-specified normalized primary.

Leave-one-piece-out analyses retained a positive estimate under all 30 single-piece omissions, with estimates ranging from 0.05225 to 0.09035. The corresponding sign-flip p-values ranged from 0.02990 to 0.11409, with median $p = 0.06549$. Thus, the estimated direction was more stable than the conventional threshold-based inferential decision.

Descriptive decomposition across all 13 states further separated raw within-piece intervention distances, the between-piece reference denominator, and the resulting normalized sensitivities. No additional layer-wise significance tests were conducted.

IV. Discussion

The intervention and RSA analyses provide different descriptive views of the same representation hierarchy. Under controlled intervention, pitch-class-versus-transposition separation generally becomes larger toward later states. Across-piece symbolic alignment, however, reaches its descriptive maximum at an earlier state. The two audit views therefore exhibit different depth profiles rather than identifying a common universally prominent layer.

The central interpretation is that *layer characterization is metric-dependent*. A layer that displays stronger selectivity under a controlled intervention need not be the layer whose across-piece geometry is most strongly aligned with an explicit symbolic descriptor. The results therefore do not support a single acoustic-to-symbolic hierarchy or a universal downstream layer-selection rule.

The post-freeze robustness audit adds a second form of measurement dependence. The sign of the primary contrast was comparatively stable, but crossing the conventional .05 threshold depended on the normalization/reference definition and on which individual work was omitted. Accordingly, normalized sensitivity is best interpreted as a conditional reference-relative audit metric rather than an absolute measure of representation quality.

V. Limitations

The conclusions are bounded to one pretrained model, one deterministic 30-work sample, one segment per work, and one synthetic rendering configuration.

The interventions are not matched for perceptual or acoustic magnitude. The piano-to-harpsichord substitution changes multiple acoustic properties simultaneously, and the tempo manipulation is confounded with clip duration. The fixed pitch-class permutation changes several aspects of pitch structure at once and should not be interpreted as a pure manipulation of symbolic organization.

Because all conditions were rendered using one FluidSynth/TimGM6mb pipeline, the observed sensitivity profiles may not generalize directly to natural recordings, alternative synthesizers, or other instrument renderings.

Whole-clip temporal mean pooling removes temporal detail and therefore makes the reported geometry dependent on the selected readout. Finally, symbolic and acoustic descriptors are explicit reference spaces, not complete definitions of symbolic musical structure or acoustics. Their RDMs are not independent conceptual axes, and RSA association does not establish causal separation between symbolic and acoustic mechanisms.

VI. Reproducibility

The primary, secondary, layer-wise, and RSA analyses were frozen before preparation of the public report. The later robustness audit is retained separately and labeled post-freeze, post-hoc, and exploratory.

The public repository provides methods, frozen result summaries, figures, robustness outputs, reusable analysis code, attribution, and reproducibility documentation. A CPU-only verifier reconstructs the formal state-12 scientific metric from the bundled verification vectors and checks exact agreement with the frozen primary outputs.

Large source audio, model caches, intermediate embeddings, and other local research artifacts are intentionally excluded from the public repository. Full model re-inference therefore requires the external dataset, model, and rendering dependencies described in the reproducibility documentation.

VII. Conclusion

This audit shows that conclusions about representation depth depend on how the representation is interrogated. Controlled intervention sensitivity and across-piece representational alignment reveal distinct descriptive depth profiles in MERT-v1-95M, while post-freeze robustness analyses show that inferential threshold crossing can also depend on the selected normalization and finite sample composition.

Rather than identifying a single "best" layer, the results motivate a measurement-aware interpretation of pretrained music representations and a clear distinction between descriptive representation profiles and confirmatory statistical claims.

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